



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences across 3,900 transactions — powered by Python, PostgreSQL, and Power BI.

3,900 Transactions, 18 Features

3,900

Total Purchases

Across all product categories

18

Data Features

Demographics, behavior & purchase details

37

Missing Values

In Review Rating column only

\$59.76

Avg. Purchase

Average transaction value (USD)

Features span customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Promo Code, Frequency, Review Rating, Shipping Type).

Python: Clean, Prepare, Load

01

Load & Explore

Imported data via pandas; used `df.info()` and `describe()` for structure and summary statistics.

02

Handle Missing Data

Imputed 37 missing Review Ratings using the median rating per product category.

03

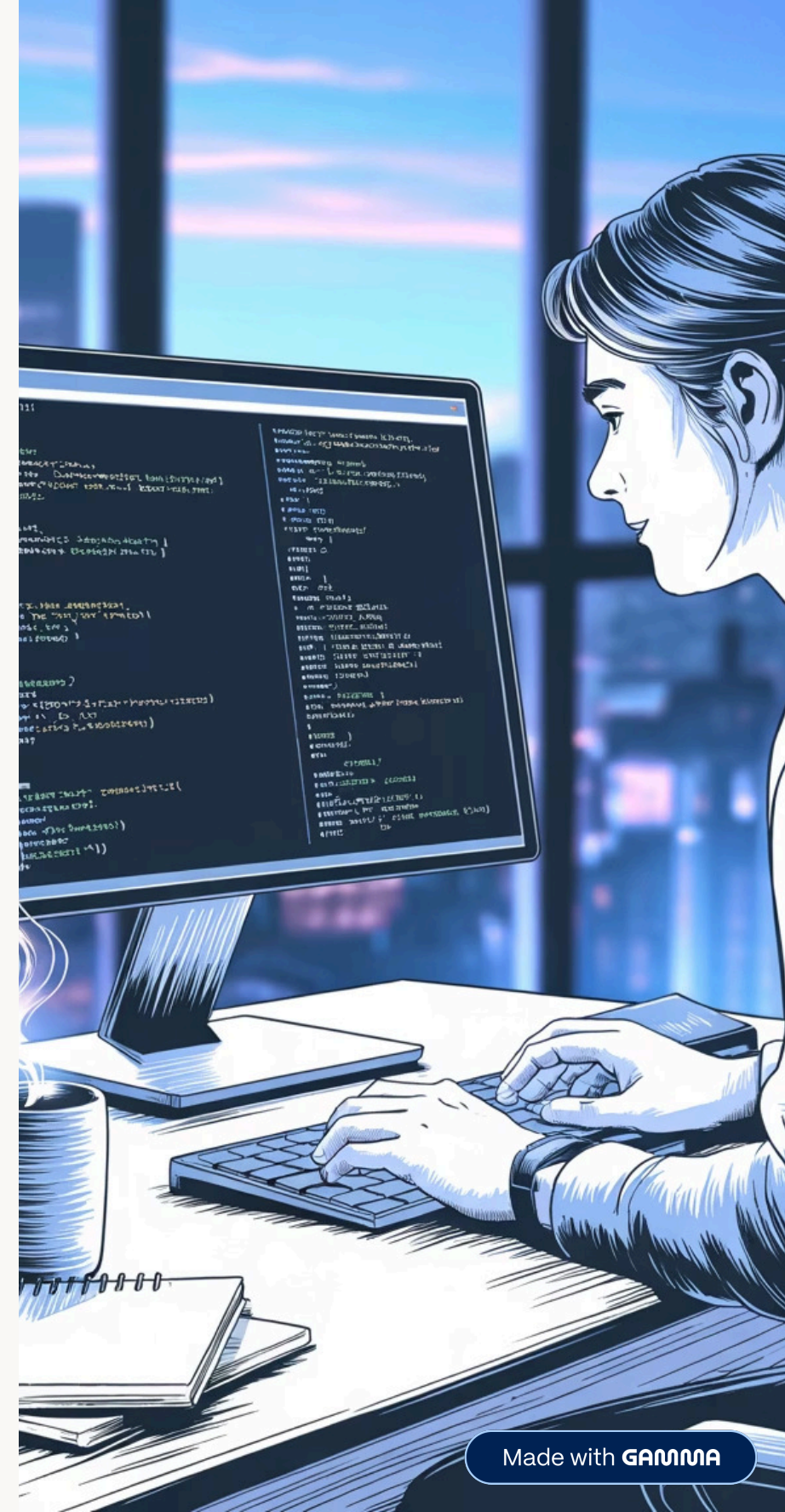
Feature Engineering

Binned ages into `age_group`; derived `purchase_frequency_days`; dropped redundant `promo_code_used`.

04

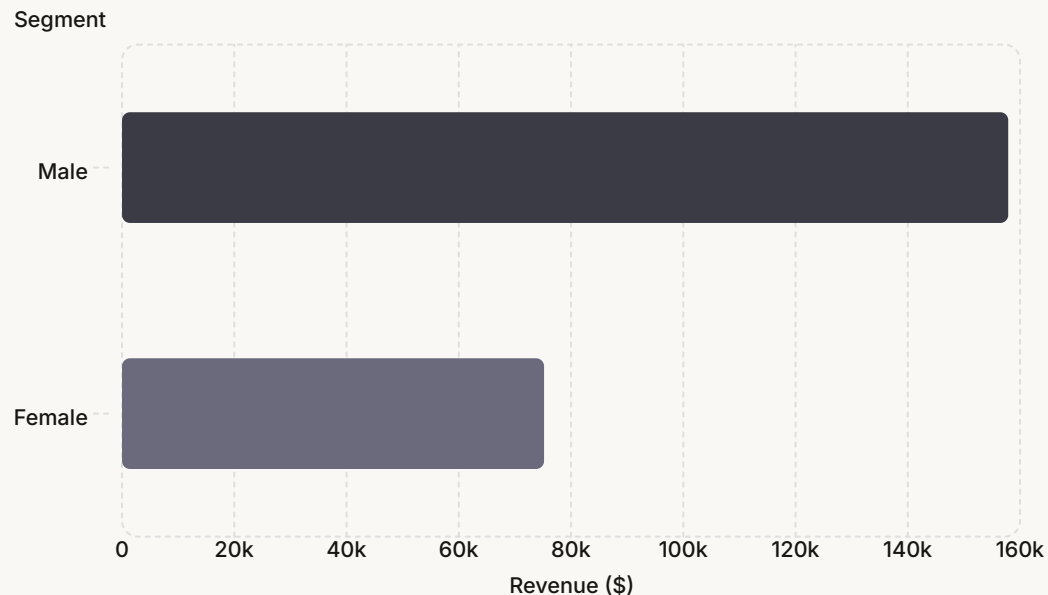
Load to PostgreSQL

Connected to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.



REVENUE INSIGHTS

Gender & Age Drive Revenue



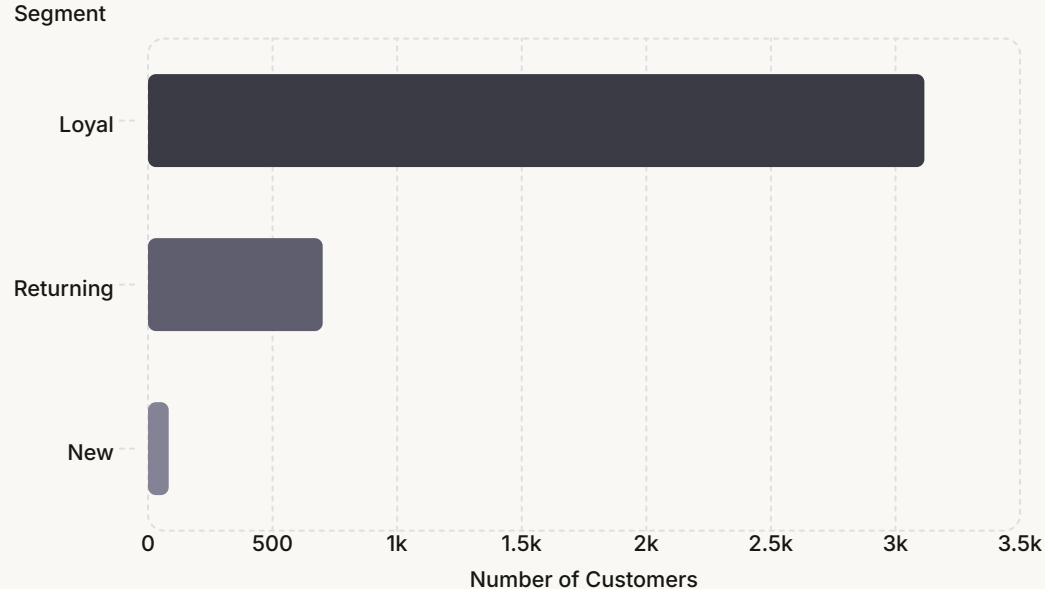
Key Findings

Male customers generated \$157,890 — more than double female revenue of \$75,191 — despite males representing 68% of all transactions.

Revenue is evenly spread across age groups: Young Adults led at \$62,143, with Middle-aged, Adult, and Senior groups each contributing \$55K–\$59K.

CUSTOMER SEGMENTATION

80% of Customers Are Loyal



Segmentation Logic

Customers classified by purchase history into **New**, **Returning**, and **Loyal** segments.

Loyal customers (3,116) dominate the base, signaling strong repeat-purchase behavior. New customers (83) represent a clear acquisition opportunity.

Of 3,476 repeat buyers, 958 hold subscriptions — a 27.6% conversion rate worth nurturing.

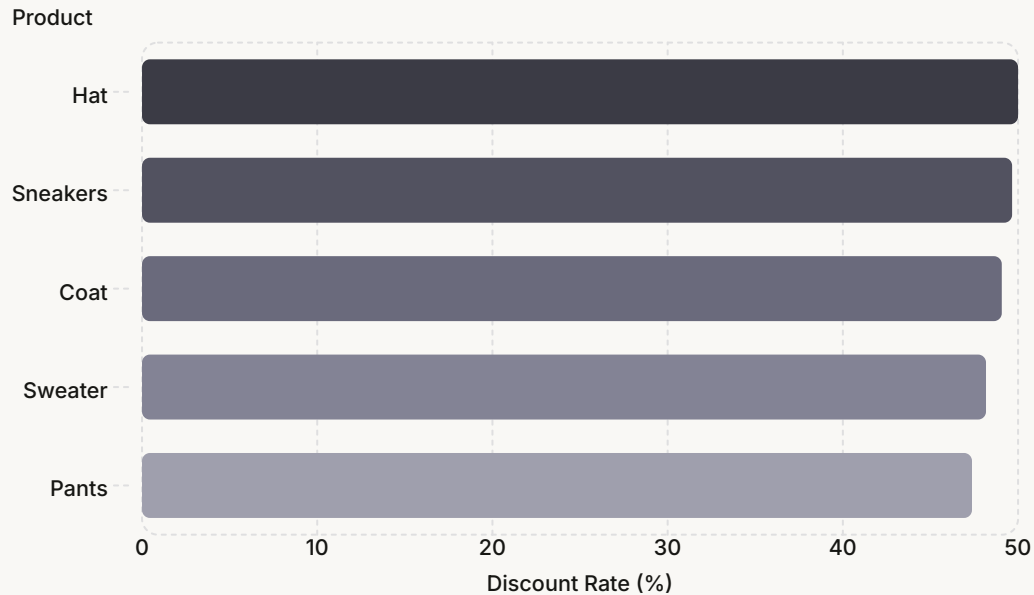
Top Products by Category & Rating

Category	Top Product	Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

Blouse and Jewelry lead their categories at 171 orders each. Gloves (3.86★) and Sandals (3.84★) top the ratings chart — strong candidates for featured promotions.

DISCOUNT BEHAVIOR

Discounts Drive Volume — But Who Really Needs Them?



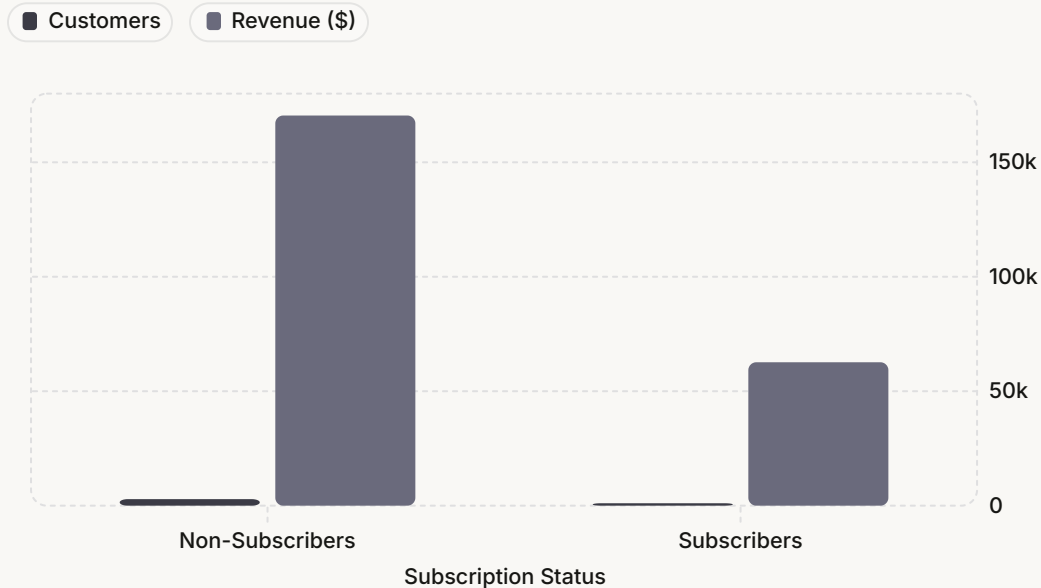
What the Data Reveals

Hats, Sneakers, and Coats are discounted in nearly **50% of all transactions** — suggesting customers may wait for promotions before purchasing.

839 discount users still spent **above the \$59.76 average**, indicating discounts may not be essential to convert these buyers. A pricing review could protect margins.

SUBSCRIPTION ANALYSIS

Subscribers Spend the Same — But There Are Fewer of Them



The Subscription Gap

Only **27% of customers** hold a subscription, yet they generate **\$62,645** in revenue — nearly a third of total sales.

Average spend is nearly identical: \$59.87 (non-subscribers) vs. \$59.49 (subscribers). The opportunity lies in converting the 2,847 non-subscribers through exclusive perks and loyalty incentives.

Interactive Dashboard: All Insights at a Glance



Dashboard Capabilities

- **KPIs:** 3.9K customers · \$59.76 avg. purchase · 3.75★ avg. rating
- **Filters:** Subscription status, Gender, Category, Shipping type
- **Charts:** Revenue & sales by category and age group
- **Drill-down:** Subscription split, shipping comparison, product rankings

Five Actions to Drive Growth



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment.



Review Discount Policy

Many discount users spend above average — reduce unnecessary promotions to protect margins.



Highlight Top Products

Feature top-rated items like Gloves and Sandals in campaigns to boost conversion.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI.